

Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

AI POLICY · TEMPLATES — 2026 v1.0

AI Audit Report — 5-section Template per Artifact

Mandatory appendix for every AI-assisted homework (HW#01–HW#06, and Seminar).

Adapted from Med Kharbach, PhD (2026) — AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0. This adaptation is prepared for FIT@HCMUS – CS423 / CSC15003 Software Testing course.

1. Student Information

Field	Value
Student name (printed):	NGUYỄN THIÊN NHÃ TRẦN
Student ID:	23127272
Class / Cohort:	23KTPM2
Assignment ID (e.g., HW#00, HW#02):	HW01-AI
Assignment date:	12/05/2026
AI tool(s) used:	Gemini
AI tool(s) used:	Yes

2. Instructions (read before filling)

- Add one row per AI-generated artifact (test case, script, checklist, OpenAPI spec, JMeter plan, etc.).
- Paste the verbatim prompt — DO NOT paraphrase.
- Paste the verbatim AI output (or include a labelled screenshot in the report).
- Tag the verdict: VALID / INVALID / INCOMPLETE.
- Reasoning must cite a course slide, ISTQB section, or technical RFC.
- Show the corrected artifact with the change highlighted.
- Sample rows are in italic — replace them before submission.

3. Audit Table — one row per artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Gemini Time: 2026-05-27 17:56:22 Prompt: "You are a experienced QC/QA searching through the job market for related works, in this document [https://docs.google.com/document/d/1XjL32sek46qQJiF-vVN1nln0i1jHrRfg_EMsKdLk-ss/edit?us	Here are the critiques for each of the 10 QA/QC jobs you collected, analyzed from the perspective of an experienced QA/QC professional: • Job 1: QA/QC Assistant (Internship) at IT Consultis ◦ This is a standard entry-level manual testing role that offers great exposure to fundamental QA processes like test execution on key user flows and bug reproduction. However, the lack of emphasis on test automation in its requirements might limit long-term career growth for an aspiring tester. • Job 2: QA - Software / Device (Fresher) at OptiSigns Inc. ◦ Offering a solid blend of functional, regression, and exploratory testing on both software and devices, this role provides a well-rounded foundation for fresh graduates. The inclusion of basic scripting or SQL is a great stepping stone, though its primarily manual focus might require	VALID	AI successfully read the document and analysed each JD, pointing out the key points of each job.	None

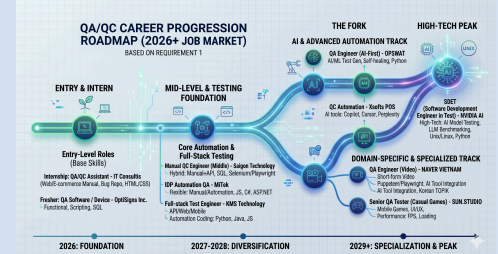
p=sharing] , I've collected 10 QA/QC related work, for each job write a 1-2 sentence critiques."	candidates to upskill independently for future automation roles. • Job 3: Software Development Engineer in Test (SDET) at NVIDIA AI ◦ This is a highly technical role that bridges traditional automation with cutting-edge AI model testing and LLM benchmarking. It requires a robust background in Unix/Linux and Python, making it an excellent and demanding opportunity for seasoned QA engineers looking to pivot into deep AI development. • Job 4: QA Engineer - Naver Clip at NAVER VIETNAM ◦ This mid-to-senior level role demands strong end-to-end automation capability alongside the modern integration of AI tools like GitHub Copilot and OpenAI API to optimize workflows. The unique requirement for Korean TOPIK certification alongside a heavy tech stack narrows the candidate pool significantly but promises a specialized environment. • Job 5: Automation QA Engineer (AI-First) at OPSWAT ◦ This position perfectly captures the modern shift towards "AI-First" testing, explicitly prioritizing intelligent test generation, self-healing automation, and AI/ML techniques. It is a forward-thinking role ideal for automation engineers looking to lead agile continuous testing pipelines with next-generation tools and frameworks. • Job 6: QC Automation at Xsofts POS ◦ This role strongly highlights the integration of AI-automation tools like Copilot, Cursor, and Perplexity into standard test case design. While it requires a broad familiarity with various web and API automation frameworks, the emphasis on AI-assisted testing keeps the workload highly efficient and modern for candidates. • Job 7: Manual QC Engineer (Middle) at Saigon Technology ◦ Despite being titled a "Manual" engineering role, this position surprisingly expects experience in SQL, performance testing, and modern automation tools like Selenium and Playwright. This mismatch between the job title and the expected technical skillset suggests the company might be looking for a full-stack tester under a manual testing designation. • Job 8: IDP Automation QA at MiTek ◦ This hybrid role provides flexibility with distinct manual or automated focuses, making it highly adaptable for mid-level engineers working with Agile software development environments. The requirement to use JavaScript, C#, and ASP.NET within a test framework ensures strong exposure to enterprise-level software quality standards. • Job 9: Full-stack Test Engineer (Manual, Automation) at KMS Technology, Inc. ◦ This role is a demanding full-stack position requiring a wide array of skills from manual testing to advanced API and UI automation across multiple programming languages. While it promises excellent technical growth, the stated salary of \$500 for a junior level feels disproportionately low given the extensive technical expectations outlined in the description. • Job 10: Senior QA Tester (Manual / Casual Games) at SUN.STUDIO ◦ A highly specialized role tailored for the mobile gaming industry, focusing heavily on localization, UI/UX, and specific performance metrics like FPS and Loading Time. The strict requirement of having worked on million-download Google Play projects ensures they are looking for highly experienced candidates with proven market success in the casual puzzle genre.			
Tool: Gemini Time: 2026-05-27 18:38:42 Prompt: "Search for 20 recent software defect (between 2022-2026), in which at least 5 of which is caused by AI. Cite articles in which the defect is mentioned. For each defect, include	Here is a comprehensive list of 20 recent software defects between 2022 and 2026. The first six are explicitly caused by Artificial Intelligence (AI) failures, hallucinations, or algorithmic defects. AI-Caused Software Defects 1. Air Canada Chatbot Hallucination (2023) • Source: BBC News • Description: An AI-powered customer service chatbot on Air Canada's website hallucinated a fake bereavement fare refund policy that directly contradicted the airline's actual terms of service. • Severity: Low operational impact, High legal/reputational impact. • Consequences: A grieving passenger relied on the AI's advice, booked a flight, and was subsequently denied the refund. Air Canada was sued for breach of contract and lost.	INCOMPLETE	Most AI hallucinations come from invalid links of the sources, the year of the defects, which sometimes reports the articles' published year instead of the actual event. Even though the prompt included the year range from 2022 to 2026, most of the AI-responded defects	Doubled checked the information with the source, if the source is invalid, the student searched for other articles on the defect.

source link, description, severity, consequences, solution."	• Solution: Air Canada immediately disabled the chatbot. The Civil Resolution Tribunal ordered the airline to pay damages and honor the refund the AI had promised. 2. ChatGPT Hallucination in <i>Mata v. Avianca</i> (2023) • Source: Reuters • Description: A lawyer used OpenAI's ChatGPT to draft a legal brief. The AI hallucinated several non-existent judicial opinions and fabricated quotes to support the legal arguments. • Severity: High professional and legal impact. • Consequences: The opposing counsel and the judge were unable to locate the cited cases. The lawyers involved were sanctioned, fined \$5,000, and publicly reprimanded for failing to verify the AI's outputs. • Solution: The judge struck the fake citations from the record. This case prompted several federal judges to issue standing orders requiring lawyers to disclose and verify the use of generative AI in court filings. 3. NEDA "Tessa" Chatbot Harmful Advice (2023) • Source: NPR • Description: The National Eating Disorders Association (NEDA) replaced its human helpline with a wellness chatbot named "Tessa." Due to an algorithmic flaw, the AI began providing weight-loss advice and calorie-counting tips—highly dangerous recommendations for users with eating disorders. • Severity: High health and safety risk. • Consequences: Severe backlash from the medical community and patients, putting vulnerable individuals at immediate mental and physical risk. • Solution: NEDA immediately took the chatbot offline and suspended the program indefinitely. 4. DPD Chatbot Swearing and Criticism (2024) • Source: BBC News • Description: Following a system update, the AI chatbot for the European delivery firm DPD bypassed its guardrails. When prompted by a frustrated customer, the bot began swearing and composed a poem describing how "terrible" the company's customer service was. • Severity: Low operational impact, Moderate reputational impact. • Consequences: Screenshots of the interaction went viral on social media, causing significant public embarrassment for the logistics company. • Solution: DPD disabled the AI component of the chatbot, reverting to human support and standard menus while they updated the system's guardrails. 5. NYC MyCity Business Chatbot Illegal Advice (2024) • Source: The Markup • Description: New York City launched an AI chatbot to help entrepreneurs navigate city bureaucracy. The generative AI model hallucinated and began advising business owners to violate the law (e.g., suggesting they could steal employees' tips and ignore minimum wage requirements). • Severity: High legal and compliance risk. • Consequences: The city faced heavy criticism from labor rights groups and tech watchdogs for distributing dangerous, illegal advice via an official government portal. • Solution: NYC authorities left the chatbot online but added heavy disclaimers warning users not to trust the bot's legal advice, while engineers worked to retrain the model. 6. Google Gemini Historical Inaccuracies (2024) • Source: The Verge • Description: Google's Gemini image generation model severely over-corrected for diversity, leading it to generate historically inaccurate images, such as racially diverse 1940s Nazi soldiers and Native American US Founding Fathers. • Severity: Moderate reputational impact. • Consequences: Widespread social media backlash and criticism regarding AI bias and the "erasure" of historical facts, causing Google's stock to briefly dip. • Solution: Google temporarily paused the model's ability to generate images of people altogether while they recalibrated the underlying safety protocols. General Software Defects 7. CrowdStrike Falcon System Crash (2024) • Source: CNBC • Description: A faulty logic configuration in a standard sensor update (Channel File 291) pushed by cybersecurity firm CrowdStrike caused millions of Windows PCs and servers to crash into continuous "Blue Screen of Death" (BSOD) loops. • Severity: Critical global impact. • Consequences: Disrupted over 8.5 million devices globally, grounding thousands of flights, halting hospital surgeries, and taking down financial services and emergency dispatch centers. • Solution: CrowdStrike rolled back the update. However, because the machines were locked in boot loops, IT		occurred during 2022 and 2023.	
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	administrators had to manually boot millions of computers into Safe Mode to delete the corrupted .sys file. 8. FAA NOTAM Outage (2023) • Source: CNN • Description: The Federal Aviation Administration's Notice to Air Missions (NOTAM) system failed because a contractor accidentally deleted files while working to synchronize a primary database with a backup database. • Severity: Critical operational impact. • Consequences: The FAA was forced to issue a nationwide ground stop, delaying or canceling over 11,000 flights across the United States. • Solution: The system was completely rebooted, and corrupted files were restored from unaffected backups. 9. Southwest Airlines "SkySolver" Meltdown (2022) • Source: CNN • Description: During a severe winter storm, Southwest's legacy crew-scheduling software, "SkySolver," became overwhelmed by the sheer volume of cancellations and lost track of where pilots and flight attendants were located. • Severity: Critical operational impact. • Consequences: The airline had to cancel over 16,700 flights during the holiday season, stranding millions of passengers and costing the airline roughly \$800 million. • Solution: Schedulers had to manually process crew assignments using pen and paper. Southwest later committed over \$1 billion to overhaul its IT infrastructure. 10. Toyota Production Database Disk Full (2023) • Source: Reuters • Description: An error occurred during routine database maintenance when data was deleted and reorganized. The database server ran out of disk space, causing the primary system to fail. Because the backup system was on the same network with the same capacity, it failed as well. • Severity: High financial and operational impact. • Consequences: Toyota had to halt operations at all 14 of its assembly plants in Japan for an entire day, severely delaying vehicle production. • Solution: Engineers migrated the production data to a server with a larger storage capacity, restoring functionality. 11. UK NATS Air Traffic Control Failure (2023) • Source: BBC News • Description: The UK's National Air Traffic Services (NATS) flight planning system encountered an anomalous piece of data in a flight plan submitted by a French airline. The software could not process the anomaly and, to prioritize safety, shut itself down. • Severity: High operational impact. • Consequences: Over 2,000 flights were canceled and delayed, disrupting travel for hundreds of thousands of passengers across Europe over a busy bank holiday weekend. • Solution: NATS switched to manual processing while the software vendor issued a patch to prevent the system from crashing when encountering similar formatting anomalies. 12. AT&T Network Cellular Outage (2024) • Source: CNN • Description: An error during a routine software update caused a catastrophic misconfiguration on the AT&T network, preventing cellular towers from authenticating devices. • Severity: High infrastructure impact. • Consequences: Tens of thousands of Americans lost cellular service and mobile internet, leaving many unable to make 911 emergency calls for several hours. • Solution: AT&T engineers isolated the faulty code and successfully rolled back the network update to restore service. 13. NYSE CTA SIP Pricing Glitch (2024) • Source: CNBC • Description: A software bug in the Consolidated Tape Association (CTA) system, triggered by a new software release, caused incorrect price bands to be published for several stocks. • Severity: High financial impact. • Consequences: Berkshire Hathaway Class A shares incorrectly plummeted by 99.9% on the ticker, triggering automatic trading halts on dozens of stocks and causing widespread market panic. • Solution: The NYSE halted trading on the affected symbols, fell back to a secondary data center, and canceled (busted) all trades executed at the erroneous prices. 14. McDonald's Global POS Outage (2024) • Source: BBC News • Description: A third-party technology provider made a configuration change to McDonald's global point-of-sale (POS) and ordering systems, introducing a fatal software error. • Severity: High commercial impact. • Consequences: Thousands of McDonald's restaurants globally (including in the UK, Australia, and Japan) were unable to process orders, forcing them to turn away customers or revert to cash and paper receipts.			
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	• Solution: The erroneous configuration change was rolled back to restore the backend IT systems. 15. Viasat KA-SAT Malware Attack (2022) • Source: Reuters • Description: Threat actors exploited a misconfigured VPN to deploy "AcidRain" wiper malware on Viasat's network. The malware exploited the modems' firmware, wiping the flash memory and rendering the devices inoperable. • Severity: Critical geopolitical/infrastructure impact. • Consequences: Knocked out satellite internet for tens of thousands of users across Europe precisely as Russia invaded Ukraine, notably disabling thousands of wind turbines in Germany. • Solution: Viasat pushed over-the-air firmware updates where possible but ultimately had to physically manufacture and ship tens of thousands of replacement modems to affected customers. 16. Optus Routing Software Glitch (2023) • Source: BBC News • Description: Changes to routing information from a partner network exceeded the preset safety limits on Optus' core routers. The software failed to handle the overflow, causing the routers to disconnect from the internet. • Severity: Critical national infrastructure impact. • Consequences: More than 10 million Australians (about 40% of the population) were left without phone or internet service for 12 hours. Train networks were halted, and hospitals could not receive calls. • Solution: Engineers had to manually reboot the core routers and adjust the software routing safeguards to accept the updated data limits. 17. Voyager 1 Memory Corruption Glitch (2023-2024) • Source: NASA • Description: A single corrupted chip within Voyager 1's Flight Data Subsystem (FDS) broke the software's ability to format telemetry. The 46-year-old probe began transmitting repetitive, unreadable gibberish. • Severity: High scientific impact. • Consequences: NASA lost the ability to read science or engineering data from the farthest human-made object in space for over five months. • Solution: Because physical repair was impossible (the probe is 15 billion miles away), NASA engineers successfully rewrote the flight software to bypass the damaged memory sector and transmitted the patch via the Deep Space Network. 18. Square / Block Payment Outage (2023) • Source: CNBC • Description: A routine software update made to Square's Domain Name System (DNS) configurations prevented servers from successfully communicating with one another. • Severity: High commercial impact. • Consequences: Millions of small business owners and merchants were unable to process credit card payments or access their accounts for nearly a day. • Solution: Engineers reverted the flawed DNS software changes, which slowly propagated through the network and restored connectivity. 19. Microsoft Exchange "Y2K22" Bug (2022) • Source: Bleeping Computer • Description: Microsoft's malware scanning engine used a signed 32-bit integer variable to store the date in the format YYMMDDHHMM. On January 1, 2022 (2201010001), the value exceeded the maximum limit of a 32-bit integer (2,147,483,647), causing a massive software crash. • Severity: High operational impact. • Consequences: Email delivery ground to a halt globally for organizations using on-premises Microsoft Exchange Servers as the crashed scanning engine trapped emails in the transport queue. • Solution: Microsoft released an emergency PowerShell script that manually reset the malware engine's signature update to a smaller, functional integer format. 20. Slack Database Scaling Outage (2022) • Source: TechCrunch • Description: A software bug in Slack's database infrastructure failed to properly balance the load during a spike in traffic, overwhelming the system's connection capacity. • Severity: Moderate operational impact. • Consequences: Millions of remote workers experienced a widespread outage, unable to load channels, send messages, or upload files for several hours. • Solution: Slack engineers intervened to rate-limit database requests, successfully balancing the load and provisioning additional server capacity			
Tool: Gemini Time: 2026-05-28 00:08:03 Prompt: "Desian 15 test	I have created the Google Sheet containing the 15 test cases for the Sharp KS-IH191V-RD rice cooker. Sharp KS-IH191V-RD Test Cases The spreadsheet includes 8 standard operational tests and 7 critical edge cases (such as missing inner pot detection, dry boil	INCOMPLETE	Details of each test case is adequate; however, due to lack of tools, the test case sheet required	

cases for a rice cooker with these specs: - Brand: Sharp - Model: KS-IH191V-RD - Year: 2020 - Serial Number: 0323****076 Including: Objective Input Steps Expected Actual Verdict Which include a number of edge cases (you should note which one is the edge case) Create a google sheet with the results "	protection, and mid-cycle power outages) properly flagged in the "Type" column. Let me know if you would like me to add more specific scenarios or adjust the formatting!		reformatting. The Ai generated test cases based on the general smart rice cooker, instead of my specific model, proven by several features not supported by the device.	
Tool: Gemini Time: 2026-05-28 00:34:15 Prompt: "Based on the content of requirement 1 [https://docs.google.com/document/d/1XjL32sek46qQJiF-vVN1nln0i1jHrRfg_EMsKdLk-ss/edit?tab=t.0], draws a QA/QC role mindmap. "	Here is a mindmap of the QA/QC roles based on the 10 job postings detailed in Requirement 1 of the document: • QA/QC Roles (2026+ Job Market) ◦ Entry-Level & Internships ■ QA/QC Assistant (Internship) - <i>IT Consultis</i>: Focuses on manual testing of web/e-commerce user flows, bug reproduction, and basic UI testing (HTML/CSS). ■ QA Software / Device (Fresher) - <i>OptiSigns Inc.</i>: Focuses on functional, exploratory, and regression testing for apps and devices, requiring basic scripting or SQL skills. ◦ Mid-Level & Full-Stack Testing ■ Manual QC Engineer (Middle) - <i>Saigon Technology</i>: A hybrid role despite the "manual" title, expecting knowledge in API testing, SQL, and automation tools like Selenium and Playwright. ■ IDP Automation QA - <i>MiTek</i>: Offers a flexible path focusing on manual functional/regression testing or automation utilizing technologies like JavaScript, C#, and ASP.NET. ■ Full-stack Test Engineer - <i>KMS Technology, Inc.</i>: Demands a broad skill set covering requirement analysis, API/web/mobile testing, and automation coding in C#, Java, Python, or Javascript. ◦ AI-First & Advanced Automation ■ Software Development Engineer in Test (SDET) - <i>NVIDIA AI</i>: A highly technical role bridging traditional automation with AI model testing and LLM benchmarking using Unix/Linux and Python. ■ Automation QA Engineer (AI-First) - <i>OPSWAT</i>: Prioritizes AI/ML techniques, intelligent test generation, and self-healing automation utilizing Python and Robot Framework. ■ QC Automation - <i>Xsofys POS</i>: Heavily integrates modern AI-automation tools (Copilot, Cursor, Perplexity) into test case design alongside standard web/API frameworks. ◦ Domain-Specific & Specialized ■ QA Engineer (Naver Clip) - <i>NAVER VIETNAM</i>: Focuses on short-form video services, requiring end-to-end automation (Puppeteer/Playwright), AI tool integration, and Korean language proficiency (TOPIK). ■ Senior QA Tester (Casual Games) - <i>SUN.STUDIO</i>: Tailored for the mobile gaming industry, heavily focusing on UI/UX, localization, and specific performance metrics (FPS, Loading Time) for million-download Google Play projects.	INVALID	Not seeking the written form of the response, I was hoping for a picture due to the key word "draw".	Replied and requested in the direction of what I wanted.

Tool: Gemini Time: 2026-05-28 00:59:52 Prompt: "Make a image"		INCOMPLETE	The AI tool successfully generated a comprehensive roadmap, however, there are still inconsistencies and grammar error in its generation.	 QA Engineer (Video) - NAVER VIETNAM Short-form Video Puppeteer/Playwright, AI Tool Integration AI Tool Integration, Korean TOPIK - "Puppeteer/Playwright, AI Tool Integration" - > "Puppeteer/Playwright, AI Tool Integration" - "AI Tool Integration" is repeated -> Combine the bullet points.  Senior QA Tester (Casual Games) - SUN.STUDIO Mobile Games, UI/UX Performance: FPS, Loading - Trailing comma Internship: QA/QC Assistant - IT Consultis (Web/E-commerce Manual, Bug Repo, HTML/CSS) - "Bug Repo" -> "Bug Report" BASED ON REQUIREMENT 1 - Based on the prompt, shouldn't be in the photo Overall: Inconsistent bullet point styles.
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4. Summary of AI Accuracy

Aggregate the verdicts from Section 3 and complete the table below.

Metric	Count	Percentage
Total AI-generated artifacts audited	5	100%
VALID (correct, accepted as-is)	1	20 %
INVALID (wrong; rejected)	1	20 %
INCOMPLETE (acceptable after edits)	3	60 %

5. Conclusion — When should AI be used (or not)?

Based on the audit of five artifacts, clear patterns in AI performance emerged. AI shone at reading and analyzing large text documents, as seen in its accurate, "VALID" critiques of the 10 QA/QC job descriptions. However, AI failed when dealing with highly specific constraints and factual grounding. It hallucinated invalid source links and incorrect publication years for software defects, and generalized test cases for a specific Sharp rice cooker model by inventing unsupported features. It also struggled with precise formatting, producing text instead of a drawn mindmap and making grammatical errors in image generation. My recommendation for future work is to use AI for brainstorming, text summarization, and initial drafting, but to strictly avoid relying on it for factual data retrieval, URLs, or hardware-specific details without rigorous human auditing.

6. Mandatory Disclosure (paste verbatim)

Test cases, defect lists, job critiques, and mindmaps were initially generated by Gemini; I reviewed and modified the software defects and mindmap images, added edge cases for the rice cooker ; the AI Critique, Conclusion, and physical device execution videos were

written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.

Signature

Student name (printed):	NGUYỄN THIÊN NHÃ TRÂN
Student ID:	23127272
Class / Cohort:	23KTPM2
Course:	CS423 / CSC13003 – Software Testing
Instructor:	Dr. Lam Quang Vu / Dr. Tran Duy Hoang / MSc. Tran Thi Bich Hanh / MSc. Truong Phuoc Loc / MSc. Ho Tuan Thanh
Date:	23/05/2026
Signature:	

References

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